

Log 708 - Mandatory 1 Template

Your Name(s)

Introduction

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

R markdown files have the suffix “.Rmd” and consist mainly of text with some formatting instructions and “code chunks” with R code.

The top few lines between --- in an .Rmd file is called a YAML header. Normally, when you start a new .Rmd file, Rstudio will set up this header for you. You can modify it, changing title and so on, but be careful to preserve spaces and other characters as is.

Below is an R “code chunk”. It is named “view_vector”. The name is optional but the {r } ... part must be there. In Rstudio, use CTRL+ALT+I to insert a new code chunk. (On a MAC: Cmd + Option + I). Then adjust it to your needs, and add code. The code below creates a vector and outputs the sum of components. Note that you can use the buttons at the top right corner of the chunk to run the code in the global environment.

My first chunk

```
z <- c(2,3,4,5,6)
z
```

```
## [1] 2 3 4 5 6
```

```
w <- data.frame(x = c(1,2,3), y = c(4,3,3))
w
```

```
##   x y
## 1 1 4
## 2 2 3
## 3 3 3
```

Then another one

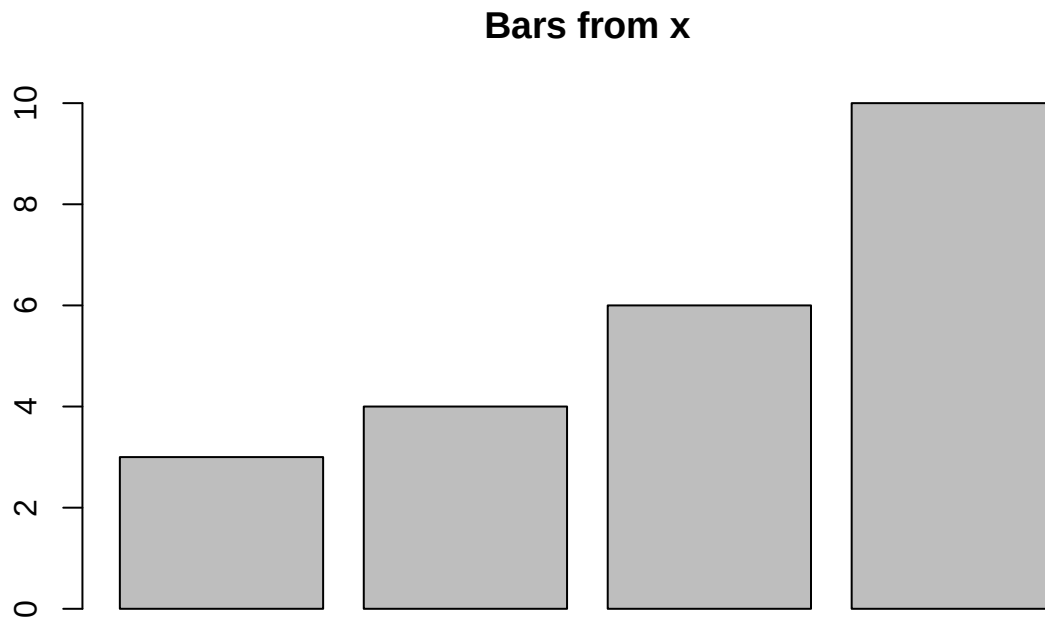
```
x <- c(3, 4, 6, 10)
sum(x)
```

```
## [1] 23
```

We may now continue to use the data created, e.g. making a plot. (Note carefully that when you “Knit” the Rmd file to html, data is created in a separate environment, *not* in the global environment. To make data available in the global environment, run the data-creating chunks first.)

Including Plots

```
#plot x and add some text labels
barplot(x, main = "Bars from x")
```



A setup for exercises

To write a solution report for exercises we can delete the above intro stuff and do something like below. The “solutions” there does not necessarily have anything to do with the actual questions. Also note that if you change your mind and don’t want to use Rmarkdown, all the code can be copied into an R script at any point. (And oppositely of course).

Exercise 1

This is our suggested solutions for exercise 1.

a. We tried this

```
#read data
#NB: you need to change the path below to make this work.
cardata <- read.csv("M:/Undervisning/Undervisningh25/Log708/Data/el_car_data_1.csv")
```

That seemed to work.

b. What variables do we have?

```
names(cardata)

## [1] "year"          "id18"          "municip_name"  "population"
## [5] "totalarea"     "income_med_adj" "pct_expense"   "sent_index"
## [9] "gasoline"      "diesel"        "el"            "hybrid"
```

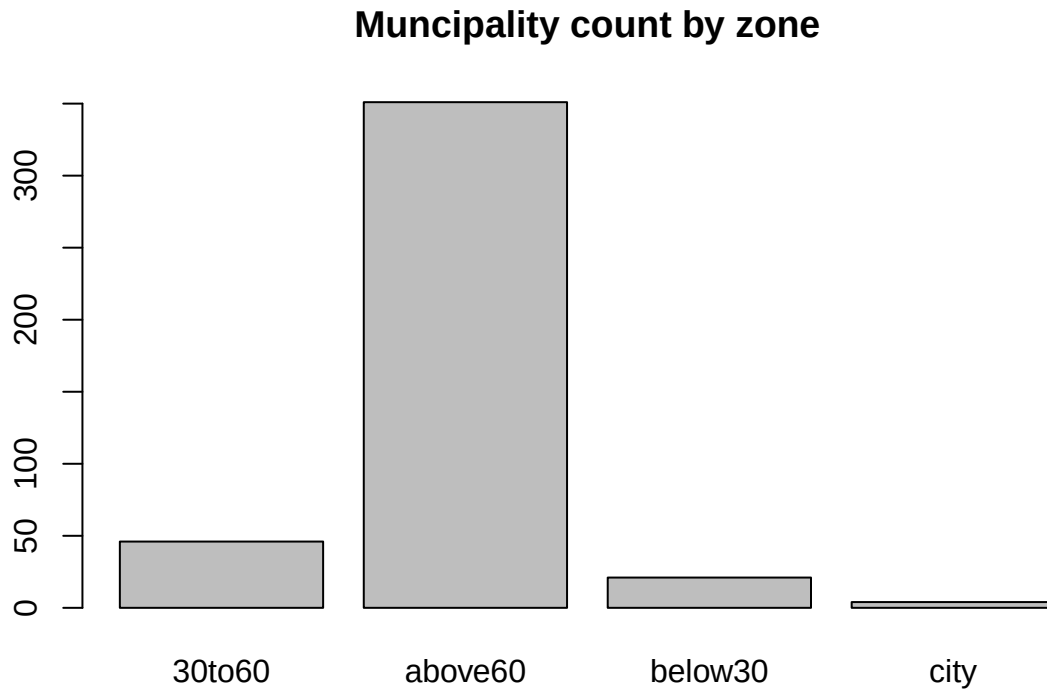
```
## [13] "plugin"          "total"          "pct_elcar"      "zone"
```

c. Make a barplot showing how many municipalities are in different zones, in 2018.

```
df18 <- subset(cardata, year == 2018)

count <- table(df18$zone)

barplot(count, main = "Municipality count by zone")
```



d. We can *hide* code in the html output, by using `echo = FALSE` in the chunk header.

```
## [1] 7 8
```

In this case, the code is evaluated, but not shown.

e. Oppositely, if we write `eval = FALSE` in the header, the code is not executed, but may be shown as below.

```
z <- c(1,2,3)
```

f. Probably, for these exercises, you won't need any of the options in d, e above, but it's ok to know them. You may want to *emphasize text* or **use boldface**. Also a bullet list can be done like

- this
- and that.
 - with sublist item
 - another item

g. One of the really nice things is the option to write maths. Expressions in text is written $x + y = \alpha + \beta$.

Use two dollar signs for single line equations, like

$$T = \frac{a-b}{a+b} = \sum_i x_i^2$$

You will find via web how to write any expression you need. The math writing is based on LaTeX.

For pdf documents you may need to do

```
install.packages("tinytex")
```

```
tinytex::install_tinytex()
```

in your Rstudio console.